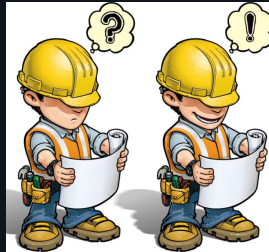


■ Constructors

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Agenda

- Recap
- Constructor
- Class vs Instances
- Marker constructor
- Glossary
- Challenge

■ Recap

1. We create a **custom (user-defined) data type**:
 - **class Marker**
2. We bind several marker **properties (attributes/instance variables)** into our Marker object:
 - **String nombreColor;**
 - **short nivelTinta;**

Marker
~nivelTinta : short ~nombreColor : String

■ Recap

What happens on line 7?

```
1 class Marker {  
2     String nombreColor;  
3     short nivelTinta;  
4 }  
5 class E11IntroPoo {  
6     public static void main(String[] args) {  
7         Marker marcadorBlanco = new Marker();  
8         marcadorBlanco.nombreColor="Blanco";  
9         marcadorBlanco.nivelTinta=100;  
10    }  
11 }
```

Constructor

Let's focus on the code after equals:

```
new Marker();
```

What does that means?

- new
- Instance?

Constructor

What does

```
();
```

means?

- We are invoking a function!
- A **special function** called **constructor**
- Did we implement any function in our marker class?

Constructor

- Did we implement any function in our marker class?

We didn't but the language provides a default constructor if we don't.

This special function handles the object creation logic.

■ Constructor

A constructor is a function **without** return value keyword and has the same name as the class:

```
1 class Marker {
2     String nombreColor;
3     short nivelTinta;
4     Marker(){
5         System.out.println("Creating a marker");
6     }
7 }
8 class E12Constructor {
9     public static void main(String[] args) {
10         new Marker();
11         Marker marcadorBlanco = new Marker();
12     }
13 }
```


Marker constructor

When we create a marker we fill out the color ink, that's part of the marker creation process.

```
1 class Marker {  
2     String nombreColor;  
3     short nivelTinta;  
4  
5     Marker(){  
6         this.nivelTinta = 100;  
7     }  
8 }
```



Glossary

constructor

A pseudo-method that creates an object. In the Java programming language, constructors are instance methods with the same name as their class. Constructors are invoked using the new keyword.

this

A Java keyword that can be used to represent an instance of the class in which it appears. this can be used to access class variables and methods.

Challenge

How can we modify the code from **E12Constructor** to assign the color String during object creation?

Hint: Use parameters



References

<https://www.oracle.com/java/technologies/glossary.html>